

PET ADOPTION & ANIMAL WELFARE MOBILE APPLICATION

PROJECT SYNOPSIS

Project Type	Final Year Project
Application Platform	Mobile Application
Frontend	Flutter / Dart
Backend Services	Firebase Authentication, Cloud Firestore, Firebase Storage
Version Control	Git / GitHub

Prepared for Final Year Project Submission

1. Introduction

The proposed **Pet Adoption & Animal Welfare Mobile Application** is a community-oriented platform designed to connect people who want to adopt pets with animals that need safe and caring homes. The application also provides a structured way for users to report stray, injured or abandoned animals and coordinate community support.

Users can register and log in, browse and search available pets, view detailed pet information, submit adoption applications and track their application status. Users can also report animals in need by providing photographs, location, condition and description. In addition, the application provides a donation/supplies section where people can offer useful items such as food, clothes, blankets and toys.

2. Problem Statement

Information about animals that need homes or immediate help is often scattered across different sources. A person may find an injured or abandoned animal but may not know how or where to report it, while shelters and animal welfare volunteers may lack a centralized platform for coordinating support.

Similarly, people willing to donate pet supplies may not know where useful food, clothes, blankets or other items are needed. The proposed application addresses these problems by bringing pet adoption, animal reporting and community supply donations into one organized mobile platform.

3. Objectives

- Provide a convenient and user-friendly mobile platform for pet adoption.
- Allow users to search and filter pets using relevant characteristics.
- Enable users to submit and track adoption applications.
- Allow people to report stray, injured or abandoned animals with photographs and location details.
- Provide a platform for users to offer useful pet supplies such as food, clothes, blankets and toys.
- Provide administrators with tools to manage pets, reports, donations and adoption requests.
- Use Firebase Authentication, Cloud Firestore, Firebase Storage and security rules to manage application data securely.

4. Major Modules and Features

User / Adopter Module: Registration and login, home screen, pet browsing, search and filters, pet details, adoption form, My Applications, application status and profile management.

Admin Module: Admin dashboard, add/edit/delete pets, upload pet images, manage pet availability, review adoption applications, approve/reject requests, review animal reports and manage donation listings.

Report an Animal: A user can upload a photograph of a stray, injured or abandoned animal along with animal type, location, condition and description. The report is stored for administrator/volunteer review. A reported animal may later be rescued, maintained as a welfare case, or added to adoption listings when appropriate.

Donate Supplies: Users can list available pet food, clothes, blankets, toys or other useful supplies, including quantity, condition, location and description. This feature is intended as a community-support coordination system rather than a full online payment/donation system.

5. Application Workflow

Adoption: Open App → Login/Register → Home → Browse/Search Pets → Pet Details → Apply → Pending → Admin Review → Approve/Reject → User receives updated status. When approved, the pet status changes from *Available* to *Adopted*.

Animal Welfare: Find Animal → Report with Photo/Location → Admin/Authorized Volunteer Review → Rescue/Action → Animal can be maintained as a welfare case or added to adoption listings when ready.

Donation: User → Donate Supplies → Enter Item Details → Submit Donation Listing → Admin/Authorized Volunteer Review → Supply can be coordinated with the relevant animal welfare organization or shelter.

6. Technology Stack

The application will use Flutter and Dart for cross-platform mobile development. Firebase services will provide authentication, cloud database and image storage.

Layer	Technology / Service
Frontend	Flutter, Dart
Authentication	Firebase Authentication
Database	Cloud Firestore
Image Storage	Firebase Storage
Development	VS Code / Android Studio
Version Control	Git, GitHub

7. Database Overview

Users: `userId`, name, email, phone and role (user/admin; volunteer role may be added if required).

Pets: `petId`, name, species, breed, age, gender, location, description, `imageUrl`, vaccinated and status.

Applications: `applicationId`, `userId`, `petId`, applicant details, reason, previous experience, date and status.

Reports: `reportId`, reporter details, animal type, image, location, condition, description, date and report status.

Donations: `donationId`, donor details, item type, quantity, condition, location, description, date and status.

8. System Architecture

The Flutter mobile application will act as the user interface. Firebase Authentication will manage user accounts and sign-in. Cloud Firestore will store user, pet, adoption application, animal report and donation data. Firebase Storage will store uploaded pet and report images. Firebase Security Rules and role-based access will help restrict sensitive operations. The Admin module will interact with these Firebase services to manage application data and workflows.

9. Flutter Project Structure

models/ – User, Pet, Application, Report, Donation and related data models.

screens/ – Authentication, user, admin and application screens.

services/ – Authentication, pet, application, report, donation and Firebase-related operations.

widgets/ – Reusable cards, buttons, text fields and other UI components.

utils/ – Constants, validation helpers and application theme.

10. Development Approach

The application will be developed in stages, beginning with Flutter and Firebase setup, followed by authentication, pet management, adoption functionality, animal reporting and donation functionality. Individual modules will first be developed and tested separately and then integrated into one complete workflow.

The final stage will focus on functional testing, validation, error handling, UI refinement, Firebase Security Rules and ensuring that the core features work reliably across supported devices.

11. Expected Outcome

The final system is expected to provide a complete community-oriented animal welfare platform. A user can discover a pet and apply for adoption, report an animal that needs help, or contribute useful supplies. Administrators can manage pets, review adoption requests, monitor animal reports and coordinate donation listings. The project will focus on completing the core workflows successfully before adding optional features.

12. Future Scope

Depending on available time and project requirements, future versions may include push notifications, favorites, location-based pet discovery, adoption history, volunteer accounts, shelter profiles, rescue-status tracking, donation matching, online monetary donations and additional community features.

13. Conclusion

The **Pet Adoption & Animal Welfare Mobile Application** aims to combine pet adoption with community-driven animal support in a single mobile platform. By using Flutter for application development and Firebase for authentication, database and storage services, the project will provide an end-to-end system for discovering pets, reporting animals in need, supporting them through useful supplies and completing the adoption process in an organized manner.